Certified Symmetry Discovery: Confidence Sets for Lie-Algebra Generators from Noisy, Limited Data

Ihor Kendiukhov[1*]

[1*]Institute of Medical Genetics and Applied Genomics, University of Tübingen, Tübingen, Germany.

Corresponding author(s). E-mail(s): kenduhov.ig@gmail.com;

Abstract

Symmetries of a dynamical system, and invariances of a trained model, are the nullspace of a linear operator: a candidate generator is a symmetry exactly when that operator annihilates it. With real data the operator is estimated, nothing is exactly zero, and a symmetry gets declared whenever some singular value looks small. That practice has no error control, and it points the wrong way: declaring a symmetry because a test failed to reject it controls the risk of missing a symmetry, not the risk of inventing one. We recast the problem as an equivalence test: a generator may be called an approximate symmetry only when the data rule out a violation larger than a stated tolerance. Two facts make this practical. Symmetry violation can be measured on a scale free of units and of the arbitrary scaling of the candidate basis, so one tolerance means the same thing across systems. And that measure is a ratio of norms of linear images of the fitted coefficients, so a confidence region propagates through it in closed form, and the bound holds for every candidate generator at once. On ten systems with exactly known symmetry algebras, across 240 settings and 72,000 runs, our procedure, SymCert, never certifies a symmetry that is not one, while a fixed threshold errs in up to 63 per cent of runs and a nullspace rank test in all of them. Noise, we find, does not manufacture false symmetries; incomplete coverage and the fitted model class do.

Keywords: Symmetry discovery, Lie algebra, Equivariance, Equivalence testing, Confidence sets, Dynamical systems

1 Introduction

Symmetry is one of the few inductive biases that reliably pays for itself in machine learning, from group-equivariant convolutions (Cohen & Welling, 2016; Kondor & Trivedi, 2018) to general constructions for arbitrary matrix groups (Finzi, Welling, & Wilson, 2021) and the geometric programme that organises them (Bronstein, Bruna, Cohen, & Veličković, 2021). All of it presumes that the symmetry is known. When it is not, it has to be discovered from data, and the question this paper is about is what a discovery may honestly claim.

A continuous symmetry of a dynamical system $\dot{x} = F(x)$ is a one-parameter family of state transformations that leaves the dynamics unchanged. Each such family is generated by a vector field ξ, and ξ generates a symmetry exactly when the Lie bracket $[\xi, F]$ vanishes identically. Because the bracket is linear in ξ, the set of generators is the nullspace of a linear operator. This observation is the engine of a large body of work: it is how symmetries of point clouds (Cahill, Mixon, & Parshall, 2023), of trained networks (Gruver, Finzi, Goldblum, & Wilson, 2023; Moskalev, Sepliarskaia, Sosnovik, & Smeulders, 2022) and of governing equations (Kaiser, Kutz, & Brunton, 2018) are found, and it was recently unified and substantially generalised by Otto, Zolman, Kutz, and Brunton (2025), who show that enforcing, discovering and promoting symmetry are all governed by the same Lie-derivative operator, with discovery reduced to computing a nullspace.

With real data the operator must be estimated, and an estimated operator has no exact nullspace. Practice therefore replaces "is this singular value zero?" with "is this singular value small?", by comparing it to a tolerance or by looking for a gap in the spectrum. This fails in two ways.

The tolerance means nothing in particular. A singular value of the Lie-derivative operator has the units of the data and depends on the arbitrary scaling of the basis chosen for the candidate generators; a tolerance of 10^{-3} is a statement about neither the system nor the evidence. Whether a small value indicates a symmetry or merely a fortunate draw of the noise depends on the noise level, the sample size, the conditioning of the design and the dimension of the model class, none of which the tolerance sees.

The burden of proof is inverted. Suppose we improve matters by attaching a hypothesis test, as a rank test would (Kleibergen & Paap, 2006; Robin & Smith, 2000): test H_0: "this generator is an exact symmetry" and declare a symmetry when the test does not reject. Such a test is valid, rejecting a true symmetry at most 5% of the time, and it is still the wrong instrument, because *failing to reject H_0 is not evidence for H_0*. The error we care about, declaring a symmetry that does not exist, is the type-II error, and no level-α test controls it. In our experiments this is not a technicality: a valid level-0.05 rank test declares a symmetry that is not one in up to 100% of replications.

Our reframing.

A procedure that may honestly say "ξ is a symmetry" must be able to *rule out* the alternative. So we ask for an *equivalence test* (Berger & Hsu, 1996; Schuirmann, 1987; Wellek, 2010): fix a tolerance δ, and certify ξ only when the data reject the hypothesis that its defect exceeds δ. This buys error control in the direction that matters: the probability of certifying a generator whose true defect exceeds δ is at most α. It also forces the tolerance into the open, where it can be argued about.

Two ingredients make it practical. First, we measure symmetry violation with a *relative* defect: the norm of the bracket $[\xi, F]$ divided by the norm of the two terms whose cancellation produces it (Definition 1). The resulting number lies in $[0, \sqrt{2}]$, is unchanged by rescaling ξ, rescaling F, or reparameterising the candidate space, and answers a concrete question: by what fraction do the two halves of the bracket fail to cancel? A tolerance stated in these units is portable across systems. Second, and this is what makes the certificate closed-form, for a model that is linear in its coefficients β, the squared defect is a *ratio of squared norms of linear images of* β (Lemma 6). A confidence ellipsoid for β therefore propagates through it exactly, by maximising and minimising a norm over an ellipsoid: two trust-region subproblems with classical closed-form solutions (Gander, Golub, & von Matt, 1989; Moré & Sorensen, 1983). Because a single event (β lies in its confidence region) implies the bound for *every* generator at once, the certificate is simultaneous over the whole candidate space, with no correction for selection or multiplicity and no sample splitting.

What we find.

We evaluate on ten polynomial dynamical systems whose symmetry algebras we compute exactly by rational linear algebra, so that every error is scored against ground truth rather than against a tolerance. Across 240 noise and sample-size settings (72,000 replications), SymCert never once certifies a generator whose true defect exceeds the tolerance, while a fixed threshold reaches 63% at its worst setting, an eigengap heuristic 100%, and rank and split-sample significance tests 100% and 92%. This error control costs detection power, and we report the cost.

Three further findings, each a consequence of taking the statistics seriously:

Noise does not manufacture symmetry; it hides it. Additive noise on the observed right-hand side biases the plug-in defect *upward*, so its effect is missed detections. The mechanisms that manufacture symmetry are incomplete coverage of the state space and, above all, the model class.

Coverage has a closed-form price. If the data occupy a region μ while the claim is about a domain ν, the defect on ν can exceed the defect measured on μ by at most an explicit factor built from two generalised eigenvalues of Gram matrices (Proposition 10); the factor is infinite exactly when the data lie in the zero set of a possible bracket, which is what happens on a single trajectory that settles onto an attractor. Across 50 coverage settings, spurious symmetry appeared only where that factor permitted it, and never in the 13 settings where it proves the contrary.

Model truncation is the dominant failure. Fitting a class too small to contain the truth produces a model with symmetries the system does not have: fitting a linear model to a cubic system yields a spurious two-dimensional algebra in 100% of replications. Sequentially thresholded least squares, the standard sparse regression of data-driven model discovery (Brunton, Proctor, & Kutz, 2016), does this automatically on a weakly symmetry-breaking system by deleting the term that breaks the symmetry. This is the one failure mode our certificate does not repair on its own, because validity is conditional on the model class containing the truth; we show that a lack-of-fit test restores control, and give a sensitivity analysis in the size of the model error for when it cannot.

Contributions.

- A scale-free relative equivariance defect (Definition 1) that makes a symmetry tolerance comparable across systems and invariant to the arbitrary choices in the nullspace formulation; minimising it is a generalised eigenvalue problem.
- SymCert: closed-form upper and lower confidence bounds on the defect of the *true* system, exact in finite samples under Gaussian errors, valid simultaneously over all candidate generators and subspaces, and hence over any data-dependent selection rule (Propositions 7 and 8). Corollaries certify approximate symmetry, refute the existence of any approximate symmetry, and report a certified symmetry dimension.
- A Le Cam lower bound (Proposition 9) showing that no procedure with false-certification rate α can certify tolerances below $c\,\sigma/\sqrt{N}$ with the constant c computable for the problem at hand; SymCert attains the same $\sigma/\sqrt{N}$ rate, within a constant factor of 8.1–8.8, of which a measured 9–18% is our own conservatism.
- A quantitative account of the three mechanisms that create apparently valid symmetries (selection optimism, incomplete coverage and model truncation), including the computable coverage factor and the demonstration that additive noise is not one of them.
- Evidence that sample splitting, the usual remedy for selection effects, is strictly counterproductive here: the simultaneous certificate is narrower than the split-sample one in all nine model configurations we tested.
- An open-source implementation, exact ground truth for every benchmark, and a full record of the compute: one pass of the thirteen experiments costs just under two CPU-hours as timed by the runner, plus a few minutes for the two experiments run outside it.

2 Setup: symmetry discovery as a nullspace problem

Generators and the Lie derivative.

Let $\dot{x} = F(x)$ be a dynamical system on $\mathbb{R}^n$ and let ξ be a vector field on $\mathbb{R}^n$. The flow of ξ is a one-parameter group of state transformations, and it maps solutions of $\dot{x} = F(x)$ to solutions if and only if the Lie bracket

$$\mathcal{L}_\xi F \;=\; [\xi, F] \;=\; DF\,\xi \;-\; D\xi\,F \tag{1}$$

vanishes identically (Olver, 1993, Ch. 2). Equation (1) is the Lie derivative in the sense of Otto et al. (2025, Eq. 16); the same expression, with $D\xi\,F$ replaced by the appropriate output action, governs symmetries of a learned map, and with $D\xi\,F$ deleted it governs invariances of a learned scalar function (Section 5.9). Everything below applies unchanged in those settings.

We search inside a *candidate class* $\mathcal{V} = \operatorname{span}\{\zeta_1, \dots, \zeta_P\}$ of vector fields, writing $\xi_\theta = \sum_p \theta_p \zeta_p$; in our experiments $\mathcal{V}$ is the set of affine generators $\xi(x) = Sx + b$, whose flows are the affine group, or all polynomial fields of degree at most two. Because (1) is linear in ξ, the exact symmetry algebra inside $\mathcal{V}$ is the nullspace $\{\theta : [\xi_\theta, F] \equiv 0\}$, and it is standard to obtain it as the nullspace of the positive semi-definite Gram operator $\langle \eta, S\xi \rangle = \int (\mathcal{L}_\eta F)^\top (\mathcal{L}_\xi F)\, d\mu$ (Otto et al., 2025, §7).

The target domain.

A symmetry is a statement about a region of state space. We therefore fix a *target measure* ν, the region over which the user wants the symmetry to hold, and define all norms as $L^2(\nu)$ norms. In our experiments ν is uniform on a box, whose monomial moments are available in closed form, so *the geometry carries no Monte-Carlo error*; all statistical uncertainty lives in the model. Keeping ν separate from the distribution μ of the observed states matters: their mismatch is the incomplete-coverage problem of Section 5.2.

The defect.

Two quantities cancel in (1). Measuring the residual relative to their size gives a dimensionless number.

Definition 1 (Relative equivariance defect)**.** For a generator ξ and a field F,

$$\rho_\nu(\xi; F) \;=\; \frac{\left\| DF\,\xi - D\xi\,F \right\|_{L^2(\nu)}}{\left(\|DF\,\xi\|^2_{L^2(\nu)} + \|D\xi\,F\|^2_{L^2(\nu)} \right)^{1/2}}. \tag{2}$$

Lemma 2 (Properties)**.** *$\rho_\nu \in [0, \sqrt{2}]$ whenever the denominator is nonzero, and $\rho_\nu(\xi; F) = 0$ if and only if ξ generates a symmetry of F on the support of ν. It is invariant under $\xi \mapsto c\xi$ and $F \mapsto c'F$ for nonzero c, c', hence under any invertible reparameterisation of the candidate class, and under similarity transformations $x \mapsto cQx$ of the state (Q orthogonal, $c > 0$) with ν transformed accordingly.*

The proof is in Section A. ρ is the fraction of the bracket's two terms that fails to cancel, so $\rho = 0.05$ means the two halves of the bracket agree to 5% in root-mean-square over the target domain. A generator with $\rho_\nu(\xi; F) \le \delta$ is what we call a *δ-approximate symmetry*; δ plays the role that the singular-value tolerance played before, but now it is comparable across systems, across candidate bases, and across units.

The denominator vanishes only if $DF\,\xi$ and $D\xi\,F$ both vanish ν-almost everywhere, which for $\xi \neq 0$ requires a degenerate F; the implementation reports any such direction as uncertified rather than assigning it a defect.

Remark 3 (Small eigenvalues are hard to read)**.** Even with a scale fixed, the smallest eigenvalue of an estimated Gram matrix is a delicate object: its fluctuations are of a different order from those of the bulk, and the classical asymptotics (Anderson, 1963) and their random-matrix refinements (Baik, Ben Arous, & Péché, 2005; Johnstone, 2001) show that the map from "small" to "zero in the population" is not one a fixed threshold can implement. This is the statistical content of the problem, and it is why the certificate below is stated for the defect ratio directly rather than for eigenvalues.

Remark 4 (An average, not a supremum)**.** ρ_ν is a mean-square quantity, so a generator can have a small defect while violating the symmetry badly on a small part of the domain, a property it shares with the Gram-operator formulation it refines and with every L^2-based criterion. Two responses are available, each with a cost. Choose ν to weight the region whose behaviour matters, which is the intended use of making ν explicit; or replace the numerator by a supremum norm, which for polynomials is bounded by a constant times the L^2 norm through a Markov-type inequality, at the cost of that constant. We use the L^2 form throughout and flag this as a modelling choice rather than a technical detail.

Remark 5 (Why the normalisation matters)**.** Without it, the "small singular value" criterion is not even well posed: rescaling one basis generator ζ_p by 10^3 rescales the corresponding row and column of the Gram matrix by 10^3 and 10^6, moving eigenvalues across any fixed threshold. ρ is a functional of the vector field ξ itself and is immune to this.

Model and data.

We take the standard measured-derivative design of data-driven model discovery (Brunton et al., 2016): pairs $(x_i, y_i)_{i=1}^N$ with $x_i \sim \mu$ and

$$y_i \;=\; F(x_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2 I_n) \text{ i.i.d.}, \tag{3}$$

and a model class $F_\beta(x) = \sum_{i,a} \beta_{i,a}\, x^a e_i$ of polynomial fields of degree at most d, so that $\beta \in \mathbb{R}^Q$ with $Q = n\binom{n+d}{d}$. Ordinary least squares gives $\hat\beta$ and, in this homoskedastic Gaussian model, an *exact* finite-sample confidence ellipsoid; Section 5.6 replaces it with a sandwich form and reports what changes. Throughout, σ is reported relative to the root-mean-square size of F over the target domain, so "noise level 0.1" means the noise standard deviation is a tenth of the typical size of the right-hand side.

Lemma 6 (Bilinear structure)**.** *For polynomial F_β and polynomial candidate class, there are matrices $L_\theta, M_\theta^{(1)}, M_\theta^{(2)} \in \mathbb{R}^{K\times Q}$, linear in θ and independent of the data, such that*

$$\rho_\nu(\xi_\theta; F_\beta)^2 \;=\; \frac{\|L_\theta \beta\|^2}{\|M_\theta^{(1)}\beta\|^2 + \|M_\theta^{(2)}\beta\|^2}, \qquad L_\theta = M_\theta^{(1)} - M_\theta^{(2)}, \tag{4}$$

where $\|\cdot\|$ is the Euclidean norm and K is the dimension of the space of possible brackets. Consequently $\min_{\theta\neq 0} \rho_\nu(\xi_\theta; F_\beta)^2$ is the smallest generalised eigenvalue of the pencil $(C(\beta), D(\beta))$ with $C = L^\top L$ and $D = M^{(1)\top} M^{(1)} + M^{(2)\top} M^{(2)}$ assembled over the basis of $\mathcal{V}$.

The matrices are obtained by expanding (1) in a monomial basis and absorbing the $L^2(\nu)$ moments into a symmetric square root; the construction is exact linear algebra and is given in Section B. The reduction to a generalised eigenproblem is itself a small improvement on the standard recipe: the plug-in symmetry directions become independent of how the candidate basis is scaled.

3 SymCert: certificates for the defect

The plug-in defect $\rho_\nu(\xi_\theta; F_{\hat\beta})$ is an estimate. What a practitioner needs is a statement about the *true* system: an upper bound on $\rho_\nu(\xi_\theta; F_{\beta^\star})$ that holds with a stated probability, for whichever generator the analysis happens to select.

3.1 The simultaneous certificate

Let $\mathcal{B} \subseteq \mathbb{R}^Q$ be any confidence region for $\beta^\star$, $\mathbb{P}(\beta^\star \in \mathcal{B}) \geq 1 - \alpha$. Define, for every $\theta \neq 0$,

$$U(\theta) \;=\; \frac{\sup\limits_{\beta \in \mathcal{B}} \|L_\theta \beta\|}{\Big(\big(\inf_{\beta\in\mathcal{B}} \|M_\theta^{(1)}\beta\|\big)^2 + \big(\inf_{\beta\in\mathcal{B}} \|M_\theta^{(2)}\beta\|\big)^2\Big)^{1/2}}, \tag{5}$$

and $L(\theta)$ by exchanging the suprema and infima.

Proposition 7 (Simultaneous validity)**.** $\mathbb{P}\big(\, L(\theta) \leq \rho_\nu(\xi_\theta; F_{\beta^\star}) \leq U(\theta) \;$ *for every* $\theta \neq 0 \,\big) \geq 1 - \alpha$. *Consequently, for* any *rule, deterministic or data-dependent, that selects a set S of generators and certifies those with $U \leq \delta$,*

$$\mathbb{P}\big(\, \exists\, \xi \in S : \; U(\xi) \leq \delta \;\; \text{and} \;\; \rho_\nu(\xi; F_{\beta^\star}) > \delta \,\big) \;\leq\; \alpha. \tag{6}$$

The proof is one line (Section A): the single event $\{\beta^\star \in \mathcal{B}\}$ implies the bound for every generator simultaneously, so the family-wise false-certification guarantee (6) comes for free. No multiplicity correction is applied because none is needed, and no data splitting is required to protect against the selection of θ, the problem that motivates split-sample and post-selection inference (Andrews, Kitagawa, & McCloskey, 2024; Berk, Brown, Buja, Zhang, & Zhao, 2013; J.D. Lee, Sun, Sun, & Taylor, 2016; Meinshausen, Meier, & Bühlmann, 2009). Section 5.4 shows empirically that splitting here is worse than doing nothing.

3.2 Closed form

Take $\mathcal{B}$ to be the ellipsoid $\{\hat\beta + \hat\Sigma^{1/2} u : \|u\| \leq R\}$.

Proposition 8 (Exact evaluation)**.** *Each supremum and infimum in (5) is the exact extremum of $\|a + Bu\|$ over the ball $\|u\| \leq R$, that is, a trust-region subproblem, solved exactly in $O(Q^3)$ by an eigendecomposition and a one-dimensional secular equation (Gander et al., 1989; Moré & Sorensen, 1983), including the degenerate "hard case". Under model (3) with ordinary least squares, taking $R^2 = Q\,\mathcal{F}_{Q,\,\mathrm{dof};\,1-\alpha}$, where $\mathcal{F}$ is* *an* F *quantile and* $\mathrm{dof} = n(N - \operatorname{rank}\Phi)$*, makes* $\mathcal{B}$ *an* exact $1-\alpha$ *confidence region in finite samples, conditional on the design.*

Using the exact trust-region extrema rather than the triangle-inequality bound $\|a\| \pm R\|B\|$ costs nothing in validity and tightens the certificate; we verify the trust-region solver against brute-force search over the ball in the test suite.

A tighter bound for a pre-specified generator.

When θ is fixed in advance, or chosen on a held-out split, the ellipsoid can be replaced by Gaussian concentration of $\|L_\theta(\hat{\beta} - \beta^\star)\|$ about its mean, giving a width driven by an *effective* dimension $\mathrm{tr}(V)/\|V\|$ with $V = L_\theta \hat{\Sigma} L_\theta^\top$ rather than by the full parameter count. Concretely, the Gaussian concentration inequality (Borell, 1975; Boucheron, Lugosi, & Massart, 2013; Cirel'son, Ibragimov, & Sudakov, 1976) applied to the 1-Lipschitz map $z \mapsto \|L_\theta \hat{\Sigma}^{1/2} z\|$, together with $\mathbb{E}\|W\| \leq (\operatorname{tr} V)^{1/2}$ and a chi-square upper bound on the noise scale, gives the valid width $(\operatorname{tr} V)^{1/2} + (2\|V\|_{\mathrm{op}} \log(1/\alpha))^{1/2}$. Section 5.4 finds that this is not in fact narrower than the simultaneous bound on our problems, so it is reported only for comparison; distribution-free alternatives such as empirical Bernstein bounds (Maurer & Pontil, 2009) could replace it where the Gaussian assumption is unwelcome.

Certify.

ξ_θ is certified as a δ-approximate symmetry when $U(\theta) \leq \delta$. Equivalently, the smallest tolerance the data support for that generator is $U(\theta)$ itself, so one may simply report U rather than choosing δ first.

Refute.

If the lower bound satisfies $\min_{\theta \neq 0} L(\theta) > \delta$ then, with confidence $1-\alpha$, the system has *no* δ-approximate symmetry in the candidate class. A bound on this minimum follows from the same ellipsoid via smallest and largest singular values (Section B).

Certified dimension.

Let V_d be spanned by the d generalised eigenvectors of $(C(\hat{\beta}), D(\hat{\beta}))$ with smallest plug-in defect. The certified dimension is the largest d such that $\sup_{\theta \in V_d} U(\theta) \leq \delta$, computed by bounding the largest and smallest singular values of the restricted maps. V_d is data-dependent, which is why Proposition 7 is stated for all θ at once.

3.3 Two ways to admit model error

Proposition 7 is conditional on $\beta^\star$ existing, i.e. on the model class containing the truth. When it does not, we replace F by $F_\beta + h$ and pay for h explicitly.

Supremum form. If $\sup_\Omega \|h\| \leq \eta_0$ and $\sup_\Omega \|Dh\|_F \leq \eta_1$ on the target domain, then $[\xi, h]$ moves the numerator by at most $\eta_1 \|\xi\|_{L^2(\nu)} + \eta_0 \|D\xi\|_{L^2(\nu)}$ and each denominator block by at most one of those terms. Valid for any smooth h; conservative, because it spends supremum norms.

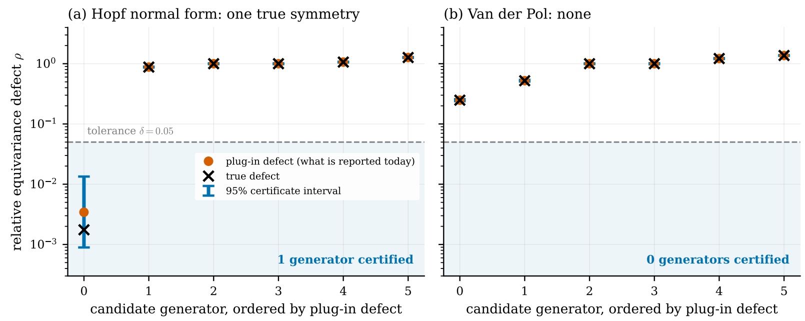

Fig. 1 What the procedure outputs, on two systems at $N = 3200$ and noise level 0.03. Each candidate generator gets an interval for its *true* defect, valid simultaneously across all six. A generator is certified when its interval lies entirely inside the shaded region below the tolerance. On the Hopf normal form the rotation generator is certified; on Van der Pol nothing is, correctly, and the lower certificate goes further, ruling out at 95% confidence the existence of *any* generator in the class with defect below 0.14. For the large defects the intervals are narrower than the plotting symbols: the certificate is only wide where the answer is genuinely uncertain.

L^2 form. If the truth lies in a *larger* polynomial class and the omitted part satisfies $\|h\|_{L^2(\nu)} \le \eta_2$, the numerator moves by at most $\|B_\theta\|_{\mathrm{op}}\,\eta_2$, where B_θ is the exact matrix of $h \mapsto [\xi_\theta, h]$ between the two $L^2(\nu)$ geometries. This is far tighter and is the recommended form. It also makes the practical advice concrete: **when in doubt, enlarge the model class**. A larger class widens the certificate, since more coefficients mean a larger ellipsoid, but does not invalidate it, whereas a class too small invalidates it silently.

Both forms are expensive because the Lie derivative *differentiates*: a model error that is small in size need not be small in slope, and it is the slope that enters the bracket. A bound stated in L^2 therefore has to be converted into a bound on a derivative, and the conversion constant grows with the degree of the class the error is allowed to live in. Stating the model-error assumption directly in a norm that controls one derivative, an H^1 bound, would remove that conversion, and is the obvious refinement; we do not pursue it here, and we report the cost of not doing so in Section 5.6.

Neither form can be verified from the data alone, so we treat η as a sensitivity parameter and report the certified tolerance as a function of it, in the manner standard for unverifiable assumptions elsewhere (Rosenbaum, 2002). In Section 5.2 we also gate certification on a lack-of-fit test, which in our experiments detects every truncated class we tried.

3.4 What is being assumed

The value of a certificate is exactly the value of its assumptions.

(A1) *Model class.* $F = F_{\beta^\star}$ for some $\beta^\star$ in the fitted class, or the deviation is bounded as in Section 3.3. This is the binding assumption; Section 5.2 is devoted to it.

(A2) *Errors.* The observed right-hand sides satisfy (3) with errors independent of the states. Exact validity needs Gaussian homoskedastic errors; a sandwich covariance (White, 1980) gives asymptotic validity otherwise. Section 5.6 measures both, and shows where the assumption genuinely fails: when the *states* are noisy and derivatives are differenced, the regressors are noisy and no covariance correction repairs the resulting bias.

(A3) *Target domain.* The claim is about the user's chosen ν. Nothing is asserted outside it. Coverage of ν by the data enters through the design, and Section 5.2 quantifies it.

(A4) *Candidate class.* We find connected subgroups whose generators lie in $\mathcal{V}$. Discrete symmetries are outside this framework, as they are outside the Lie-algebraic framework generally.

4 How small a violation can anyone certify?

Two questions bound what any procedure can do. How small a tolerance is achievable, and when can the observed region of state space stand in for the target domain?

4.1 A resolution limit

Exact symmetry cannot be certified from noisy data: a system with zero defect and one with a defect just above zero generate almost the same data. The useful version of this statement is quantitative.

Proposition 9 (Le Cam bound on the certifiable tolerance)**.** *Consider model* (3) *with fixed design. Let $\varphi \in \{0,1\}$ be any rule for certifying a fixed generator ξ at tolerance δ whose false-certification rate is at most α, i.e. $\mathbb{P}_\beta(\varphi = 1) \le \alpha$ whenever $\rho_\nu(\xi; F_\beta) > \delta$. Then for any β^0 with $\rho_\nu(\xi; F_{\beta^0}) = 0$,*

$$\mathbb{P}_{\beta^0}(\varphi = 1) \;\le\; \alpha \;+\; \frac{\delta\sqrt{N}\,\|M_\xi \beta^0\|}{2\,\sigma\, s_{\max}(\xi)} \;+\; o(\delta), \qquad s_{\max}(\xi) = \max_{\Delta \neq 0} \frac{\|L_\xi \Delta\|}{\|F_\Delta\|_{L^2(\mu)}}. \tag{7}$$

Equivalently, attaining power γ requires $\delta \gtrsim 2\sigma\, s_{\max}(\xi)\,(\gamma - \alpha)\,/\,(\|M_\xi\beta^0\|\sqrt{N})$.

The constants are computable for a given problem, so this is a numerical benchmark rather than a rate statement. Section 5.3 evaluates it: SymCert's certified tolerance tracks the bound's $\sigma/\sqrt{N}$ scaling over two decades of N and three noise levels, exceeding it by a factor of 8.1 to 8.8. Part of that gap is ours, since a calibration-targeting bootstrap is 9–18% narrower (Section 5.5); the rest is not attributable without a matching construction: Proposition 9 is a lower bound obtained through Pinsker's inequality and a single two-point alternative, and we do not claim it is attained. What the comparison does establish is that the *rate* is right: no procedure, however clever, can improve on SymCert by more than a constant factor.

4.2 The price of incomplete coverage

Suppose the data occupy a region with distribution μ while the claim is about ν. Every bracket $[\xi_\theta, F_\beta]$ lies in a fixed finite-dimensional space $\mathcal{R}$ of polynomial fields, so the worst-case discrepancy between the two measures is a single generalised eigenvalue.

Proposition 10 (Extrapolation factor)**.** *Let G_μ and G_ν be the Gram matrices of a basis of $\mathcal{R}$ under μ and ν, and set $\kappa = \lambda_{\max}(G_\mu^{-1/2} G_\nu G_\mu^{-1/2})$ and $\kappa' = \lambda_{\max}(G_\nu^{-1/2} G_\mu G_\nu^{-1/2})$. Then for every $r \in \mathcal{R}$, $\|r\|^2_{L^2(\nu)} \le \kappa \|r\|^2_{L^2(\mu)}$, with equality attained, and consequently*

$$\rho_\nu(\xi; F) \;\le\; \sqrt{\kappa\kappa'}\, \rho_\mu(\xi; F) \qquad \text{for every } \xi \in \mathcal{V}. \tag{8}$$

Moreover $\kappa = \infty$ if and only if some nonzero $r \in \mathcal{R}$ vanishes μ-almost everywhere, i.e. the data lie in the zero set of a possible bracket.

As a *diagnostic*, κ and κ' are computable from the observed states alone, before any symmetry analysis, and $\sqrt{\kappa\kappa'}\,\delta$ is the tolerance that a defect of δ measured on the data can actually support on the target domain. As an *explanation*: declaring a symmetry at tolerance δ from a defect measured on the data requires some generator whose *true* defect satisfies $\rho_\nu \le \sqrt{\kappa\kappa'}\,\delta$. Writing $\rho^\star$ for the smallest true defect in the candidate class that is not already within tolerance, a spurious symmetry is possible only if

$$\sqrt{\kappa\kappa'} \;\ge\; \rho^\star/\delta. \tag{9}$$

This is a one-sided guarantee. Falling below the threshold *proves* that no spurious symmetry can occur; exceeding it is a warning rather than a prediction, since the bound is worst-case over the residual space. Section 5.2 confirms both halves: no false certification anywhere below the threshold, and false certifications on some but not all settings above it.

The infinite case is not exotic. A single trajectory of a planar system with a limit cycle eventually lies on an algebraic curve, and every bracket divisible by the curve's defining polynomial vanishes on the data. We observe exactly this (Table 2).

4.3 Why noise alone does not manufacture symmetry

It is natural to expect that noise makes small singular values smaller. Under model (3) the opposite happens. Since $\mathbb{E}\|L_\theta\hat\beta\|^2 = \|L_\theta\beta^\star\|^2 + \mathrm{tr}(L_\theta \Sigma L_\theta^\top)$, and likewise for the denominator, estimation error inflates *both* parts of (4); for a direction that is close to a true symmetry the numerator is near zero, so its relative inflation is unbounded while the denominator's is not. The plug-in defect of a true symmetry is therefore biased upward and all directions are pulled toward a common noise-dominated value. The practical consequence is that a fixed tolerance loses *detections* as the noise grows, at a rate that depends on σ, N and the design, none of which the tolerance knows, rather than gaining false ones. Section 5.1 measures this over 32,000 replications spanning noise levels from 0.1 to 1.0 and sample sizes from 30 to 400 on four systems with no symmetry: the median smallest plug-in defect is 1.01, 1.09, 1.27 and 1.64 times the true one at the four noise levels, and in not one of the 32,000 replications did it fall below 0.05.

A second effect pulls the other way. The generator chosen to *minimise* the plug-in defect is selected for a small value, so its true defect tends to exceed its plug-in value: the winner's curse. The two effects compete, and which wins is system-dependent:

at noise level 0.2 and $N = 100$ the mean of $\rho_\nu(\hat{\xi}; \beta^\star) - \hat{\rho}(\hat{\xi})$ is $+0.0150 \pm 0.0017$ on Van der Pol, $+0.0078 \pm 0.0004$ on Lotka–Volterra, $+0.0060 \pm 0.0021$ on Duffing and -0.0073 ± 0.0009 on the asymmetric rigid body. Both effects decay with N (the Van der Pol gap falls to 0.0002 ± 0.0002 at $N = 6400$), and neither is large. But a selection effect of either sign is enough to invalidate a test evaluated on the data that chose the direction, and Proposition 7 removes the need to reason about it at all.

5 Experiments

Design.

Ten polynomial systems on $\mathbb{R}^2$ and $\mathbb{R}^3$ serve as benchmarks: four with a nonzero symmetry algebra inside the affine candidate class (a Hopf normal form, a planar rotationally symmetric linear system, the Euler equations of a symmetric top, and a radial flow onto the sphere with full $\mathfrak{so}(3)$), and six with none (Van der Pol, Duffing, Lotka–Volterra, Sel'kov, an asymmetric rigid body, and Lorenz). Every system has rational coefficients, so its exact symmetry algebra is the exact rational nullspace of the bracket map, computed with symbolic arithmetic; the true defect of any generator is likewise available in closed form from monomial moments. **Ground truth involves no tolerance and no sampling**, which matters here because the quantity being estimated is itself a near-degeneracy. The exact algebra dimensions are cross-checked against an independent characterisation (for linear systems, the dimension of the commutant of the coefficient matrix).

We measure two things, identically for every method. *False certification* is the event that the declared subspace contains a generator whose true defect exceeds δ; because the declared object is a subspace, this is evaluated as a generalised eigenvalue of the true forms restricted to it, not by sampling directions. *Detection* is the event that the declared dimension is at least the true one. Unless stated otherwise $\delta = \alpha = 0.05$.

Methods compared.

The framework we build on does not prescribe a tolerance (it establishes that the symmetries are a nullspace), so there is no single baseline to compare against, and we sweep the tolerance instead of picking one. We also give the threshold rule the advantage of the *exact* target-domain Gram matrix in Section 5.1, which no practitioner without our construction would have; that isolates the purely statistical question and removes the coverage effect, which is studied separately in Section 5.2. The methods are (i) a fixed threshold on the plug-in defect, swept over $\tau \in [0.005, 0.4]$; (ii) the largest relative eigengap; (iii) a Wald rank test for the nullspace dimension in the spirit of Robin and Smith (2000) and Kleibergen and Paap (2006); (iv) a split-sample significance test that selects a direction on one half and tests H_0: exact symmetry on the other, declaring a symmetry when it fails to reject; (v) a Weyl-inequality certificate, the natural singular-value perturbation answer; (vi) SymCert; (vii) SymCert with sample splitting and the tighter pointwise bound. Full definitions are in Section C.

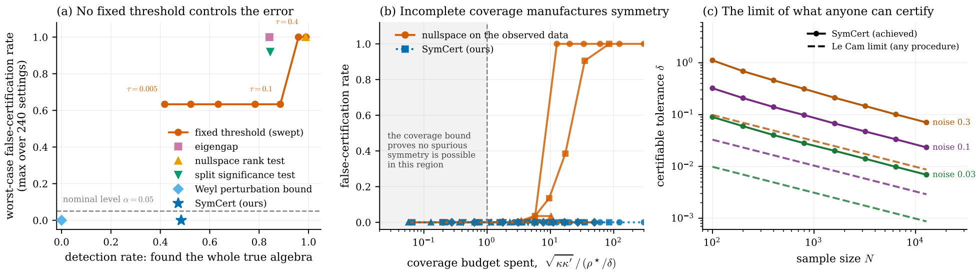

Fig. 2 **(a)** Every method is placed by how often it finds the whole true symmetry algebra (horizontal) against the largest false-certification rate it produces over the 240 settings (vertical). The orange curve traces the fixed-threshold rule as the tolerance is swept from 0.005 to 0.4: no choice of tolerance brings its worst case near the nominal level (dashed). The rank test and the split-sample significance test are valid level-0.05 tests, and are among the worst discovery rules, because failing to reject "this is a symmetry" is read as evidence for it. SymCert is at 0; the Weyl perturbation bound is also at 0 but never certifies anything. **(b)** Shrinking the region the data occupy while keeping the claim about the full domain. The horizontal axis is the extrapolation bound of Proposition 10 divided by the value it must reach before a spurious symmetry is arithmetically possible. In the shaded region the proposition *proves* no spurious symmetry can occur, and none does in any of the 13 settings there; beyond it the nullspace rule fails on 14 of 37 settings. SymCert (blue) is at zero throughout, declining to certify rather than certifying wrongly. **(c)** The tolerance that can be certified about a truly symmetric system, against the Le Cam limit of Proposition 9 for any procedure with the same error control. Both scale as $\sigma/\sqrt{N}$; the gap is a constant factor of 8.1–8.8.

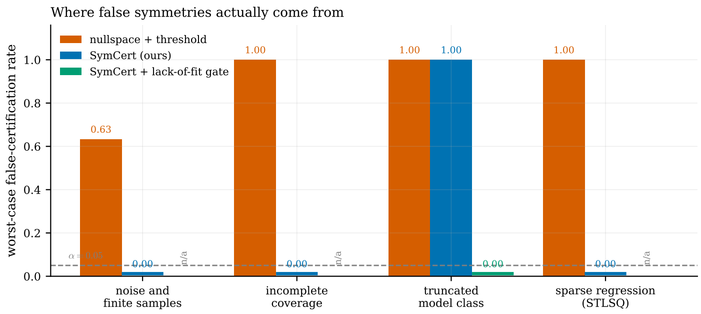

Fig. 3 Worst-case false-certification rate under each of the four candidate mechanisms, at tolerance $\delta = 0.05$. Noise and finite samples alone are the mildest; incomplete coverage and, above all, the model class are what manufacture symmetry. SymCert controls the first two by construction. It does *not* control the third, because its validity is conditional on the model class containing the truth; a lack-of-fit test detects every truncated class we tried and restores control (green). The fourth group is sparse regression on a weakly symmetry-breaking system: the sparsity threshold deletes the term that breaks the symmetry, and the resulting model is exactly symmetric. Certifying from the unpruned fit gives a rate of zero.

5.1 Calibration and power

Figure 2(a) summarises 240 settings (10 systems × 6 sample sizes × 4 noise levels, 300 replications each; 72,000 replications in total).

No fixed threshold is safe. Lowering τ does not fix the problem: even at $\tau = 0.005$ the worst setting has a 63% false-certification rate, while detection has already fallen

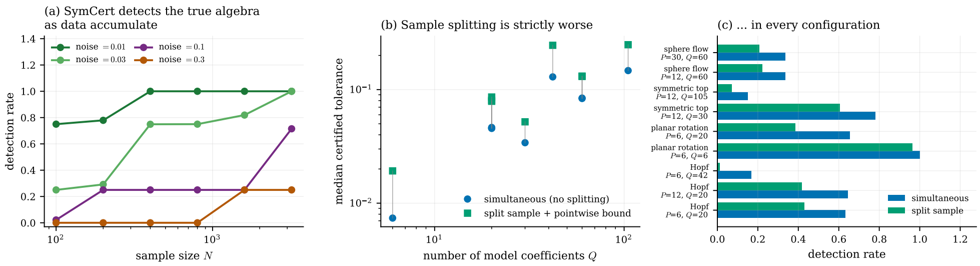

Fig. 4 **(a)** The price of validity is detection power, and it is paid in sample size: at noise level 0.01 SymCert recovers the full true algebra in every replication from $N = 400$; at noise level 0.3 it does not, because the data do not support the claim. **(b,c)** Simultaneity versus sample splitting. Each point in (b) is one model configuration; the simultaneous certificate (blue) is narrower than the split-sample certificate (green) in all nine, by factors of 1.5 to 2.6, and (c) its detection rate is higher in all nine. Splitting protects against a selection effect that simultaneity already removes, and pays for that protection with half the data.

	false certification		detection	
method	mean	worst setting	mean	settings above α
fixed threshold $\tau = 0.005$	0.005	0.633	0.418	3 / 240
fixed threshold $\tau = 0.01$	0.005	0.633	0.523	3 / 240
fixed threshold $\tau = 0.02$	0.005	0.633	0.635	3 / 240
fixed threshold $\tau = 0.05$	0.005	0.633	0.784	3 / 240
fixed threshold $\tau = 0.1$	0.017	0.633	0.886	18 / 240
fixed threshold $\tau = 0.2$	0.139	1.000	0.960	52 / 240
fixed threshold $\tau = 0.4$	0.551	1.000	0.990	142 / 240
largest eigengap	0.650	1.000	0.842	166 / 240
nullspace rank test	0.070	1.000	0.988	30 / 240
split significance test	0.070	0.920	0.845	37 / 240
Weyl perturbation bound	0.000	0.000	0.000	0 / 240
SymCert (ours)	**0.000**	**0.000**	0.484	**0 / 240**
SymCert + sample splitting	**0.000**	**0.000**	0.370	**0 / 240**

Table 1 Calibration and power over 240 settings, 300 replications each. "Worst setting" is the maximum false-certification rate over the settings; the last column counts those exceeding the nominal $\alpha = 0.05$. No fixed threshold controls the error, and the two significance-test procedures — both valid tests — are the worst offenders relative to their apparent rigour. The Weyl certificate is valid but never certifies anything.

to 0.42. The offending settings are the *symmetric* systems at high noise, where the threshold recovers the right *dimension* but the wrong *subspace*: the selected eigenvectors have rotated away from the true algebra. A metric that only counted dimensions would miss this entirely.

Valid tests are not safe either. The rank test and the split-sample significance test are correct level-0.05 tests of "this is an exact symmetry". Used as discovery rules they reach false-certification rates of 1.00 and 0.92. This is the inverted burden of proof

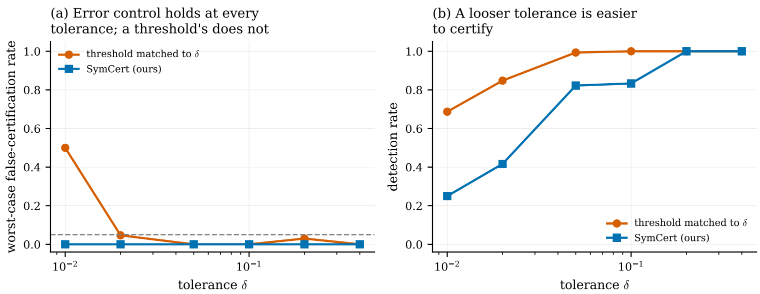

Fig. 5 Sweeping the tolerance, on a smaller grid: six systems, two sample sizes, two noise levels. **(a)** SymCert's worst-case false-certification rate is zero at every tolerance; the threshold rule matched to the same δ exceeds the nominal level at the tightest one and happens to stay below it at the rest; on the full grid of Table 1 it does not. **(b)** The tolerance is a real choice rather than a nuisance parameter: it is the approximation the user is willing to accept, and accepting more of it makes symmetry easier to certify, from a detection rate of 0.25 at $\delta = 0.01$ to 1.00 at $\delta = 0.2$.

made quantitative: their power is lowest exactly where the data are least informative, and low power is read as evidence for symmetry.

The tolerance is a choice, and it changes what is at stake. Figure 5 repeats the comparison across $\delta \in [0.01, 0.4]$ on a smaller grid of six systems, two sample sizes and two noise levels. SymCert's worst-case error is zero at every tolerance, and its detection rate climbs from 0.25 at $\delta = 0.01$ to 1.00 at $\delta = 0.2$: a user who will accept a 20% relative defect can have both error control and full detection here, and one who insists on 1% cannot. The threshold rule's worst case is 0.50 at $\delta = 0.01$ and below the nominal level at the other five values. That is not a defence of it. On this smaller and easier grid it happens to be safe at most tolerances; on the full grid of Table 1 its worst case is 0.63 at every τ we tried. Which of those two situations one is in is precisely what a threshold cannot tell you, and what the certificate can.

Validity is not free, and neither is it vacuous. SymCert never errs, at a detection cost relative to a threshold at the same tolerance (0.48 against 0.78). Figure 4(a) shows where the cost falls: at noise level 0.01 detection reaches 1.00 by $N = 400$; at noise level 0.30 it does not, because, correctly, the data at that noise level do not support the claim. The Weyl certificate shows the alternative failure: valid, and useless.

5.2 Noise, coverage, and the model class

Figure 3 isolates the four candidate mechanisms.

Noise is not the culprit.

With the model class correct and the data covering the target domain, the threshold rule's errors are almost all missed detections. We push this deliberately hard: four systems with no symmetry, noise levels from 0.1 to 1.0, sample sizes from 30 to 400, 500 replications per cell, 32,000 in all. The median smallest plug-in defect *rises* with the noise, to 1.64 times its true value at noise level 1.0, and across all 32,000 replications the smallest value ever observed was 0.066, still above the tolerance. This confirms the upward bias argued in Section 4.3: additive noise on the right-hand side inflates the

system	design	points	$\sqrt{\kappa\kappa'}$	true dim.	nullspace dim.	SymCert dim.
Van der Pol	single trajectory	3000	34.9	0	0	0
Van der Pol	8 trajectories	3200	9.7	0	0	0
Van der Pol	uniform design	3000	1.1	0	0	0
Hopf	single trajectory	3000	∞	1	1	0
Hopf	8 trajectories	3200	24.0	1	1	1
Hopf	uniform design	3000	1.1	1	1	1
Lorenz	single trajectory	3000	231.3	0	0	0
Lorenz	8 trajectories	3200	84.4	0	0	0
Lorenz	uniform design	3000	1.1	0	0	0

Table 2 Trajectory data. A single trajectory that settles onto an attractor gives an infinite or large extrapolation factor: the data lie in, or near, the zero set of a possible bracket, so a defect measured on them says little about the target domain. SymCert declines to certify exactly there.

numerator of the defect relative to a true symmetry, so it removes symmetries rather than inventing them.

Coverage.

We shrink the region the data occupy from the full target box to a box of 0.15 times the half-width, keeping $N = 2000$ and noise level 0.02, and run the nullspace computation on the empirical measure of the data, which is what one does when the sample is taken to stand for the domain of interest. Figure 2(b) plots the false-certification rate against the extrapolation bound $\sqrt{\kappa\kappa'}$ divided by the value $\rho^\star/\delta$ it must reach for a spurious symmetry to be possible at all, from (9). Of the 50 settings, 13 fall in the region where Proposition 10 *proves* that no spurious symmetry can occur, and in all 13 the observed rate is exactly zero; of the 37 beyond it, 14 do produce false certifications, rising to a rate of 1.0. SymCert, which measures the defect on the target domain and inherits the design's uncertainty through $\hat{\Sigma}$, never certifies falsely and instead reduces its certified dimension to zero as coverage degrades: on the Hopf system it certifies the true one-dimensional algebra when the data cover the domain and certifies nothing once the data occupy less than half of it.

Table 2 shows what the diagnostic says about trajectory data, the way such data actually arrive. A single trajectory of the Hopf system converges to the unit circle, an algebraic curve, so some bracket vanishes on the data exactly and the extrapolation factor is infinite: a defect measured on those points constrains nothing on the target domain, and any tolerance whatever can be met there by a generator that is not a symmetry. Eight trajectories bring the factor down to 24 and a uniform design to 1.1. In these particular runs the nullspace rule happens to give the right dimension on every row (these are single realisations, not rates), but on the single Hopf trajectory it does so without support from the data, and SymCert correctly declines. The corresponding rates, and the failures, are in Figure 2(b).

The model class dominates.

Fitting a class too small to contain the truth is the strongest mechanism we found, and the only one that defeats SymCert as well. Fitting a linear or quadratic model to a cubic system produces a fitted field with symmetries the system does not have, since a linear system always has an n-dimensional symmetry algebra, and both the threshold rule and SymCert certify them in up to 100% of replications (Figure 3, third group). This is a failure of assumption (A1), not of the certificate. A lack-of-fit F-test against the correct class rejects in 100% of replications in every truncated setting we ran; gating certification on it returns the false-certification rate to zero without affecting the correctly specified settings.

The same failure arises without anyone choosing to truncate. Sequentially thresholded least squares, the sparse regression at the core of data-driven model discovery (Brunton et al., 2016; Champion, Zheng, Aravkin, Brunton, & Kutz, 2020; Kaptanoglu et al., 2022) and the object of its own convergence theory (Zhang & Schaeffer, 2019), deletes small coefficients. On a Hopf normal form perturbed by a symmetry-breaking term of size ε, whose rotation defect is 0.064 at $\varepsilon = 0.1$, above the tolerance, a sparsity threshold of 0.25 deletes the breaking term and reports an exact rotational symmetry in 100% of replications, against 17% with no thresholding. Certifying from the unpruned least-squares fit gives zero. Sparsification and symmetry discovery interact, and the interaction is invisible if the two steps are run in sequence without accounting for the first.

5.3 How small a violation can be certified

Figure 2(c) tracks the certified tolerance on the one-parameter family F_ε = Hopf + $\varepsilon x^2 \partial_x$, whose rotation defect grows continuously from zero. Over two decades of N and three noise levels the certified tolerance follows the $\sigma/\sqrt{N}$ scaling of the Le Cam bound of Proposition 9, exceeding it by a factor 8.1, 8.3 and 8.8 at noise levels 0.03, 0.1 and 0.3. The factor is essentially constant, so the achieved rate is optimal and only the constant is lost. Section 5.5 measures the part of that constant that is ours rather than the bound's.

The same experiment measures the opposite claim. At $N = 1600$ and noise level 0.1, the lower certificate rules out exact symmetry ($L > 0$) for 49.5% of replications when the true rotation defect is 0.013 and for 100% when it is 0.032; when the system really is symmetric it does so in 1.0% of replications, against the nominal 5%.

5.4 Problem dimension, and whether to split the sample

The width of the simultaneous certificate is governed by the number of model coefficients Q, through the radius of the confidence ellipsoid; the width of the split-sample pointwise bound is governed by an effective dimension of the generator being tested, but it sees half the data. Figure 4(b,c) settles which wins: **the simultaneous certificate is narrower in all nine configurations**, by factors of 1.4 to 2.6, and its detection rate is higher in all nine. Sample splitting, the standard remedy for the

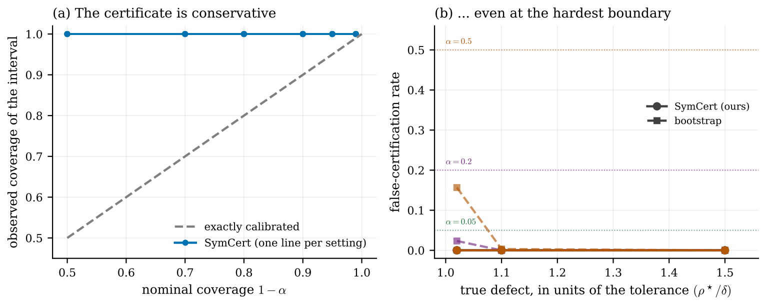

Fig. 6 **(a)** Observed coverage of the certificate interval against its nominal level, one line per setting. The certificate over-covers: the bound treats numerator and denominator separately over the whole confidence ellipsoid. **(b)** The deliberately hardest case: a system whose true defect sits just above the tolerance, observed at the sample size that puts the median certified bound exactly at the tolerance. SymCert (solid) never certifies falsely; a calibration-targeting bootstrap (dashed), which has no finite-sample guarantee, does. Colour marks the nominal level, matching the dotted reference lines.

selection effect and one of the two most often suggested for this problem, is counterproductive here, because the selection effect it protects against is already handled for free by simultaneity.

The reason is visible in Section 5.3's data. For a generator fixed in advance, where splitting is unnecessary, the pointwise bound is never more than 5% narrower than the simultaneous one, and at small samples it is up to 1.9 times *wider*, because the Gaussian-concentration bound it rests on is loose there. Simultaneity over the entire candidate space is therefore essentially free for this functional, while splitting costs the usual $\sqrt{2}$ in effective sample size.

Two further readings. Increasing the *candidate generator* class is nearly free: moving from the 6-dimensional affine class to the 12-dimensional class of all quadratic generators on the Hopf system changes the median certified tolerance from 0.0464 to 0.0457. Increasing the *model* class is not: raising the polynomial degree from 3 to 5 (Q from 20 to 42) widens it from 0.046 to 0.129. The asymmetry is useful: search widely over generators, and spend your sample-size budget on the model class.

5.5 Is the guarantee tight?

A certificate that certified nothing would also never certify anything false. Figure 6 shows that ours is conservative but not vacuous. Panel (a): the interval covers the true defect more often than its nominal level requires, the observed coverage being 1.000 at every nominal level from 0.50 to 0.99, because (5) bounds numerator and denominator separately over the whole ellipsoid. Panel (b) constructs the worst case deliberately: a system whose true defect is $(1+\tau)\delta$ for τ as small as 0.02, observed at the sample size that puts the median certified bound exactly at δ. Across nine such configurations and 2,700 replications the false-certification rate is 0.

To quantify the slack, we compare against a calibration-*targeting* alternative in the spirit of the parametric bootstrap (Efron, 1979): draw $\beta^* \sim N(\hat{\beta}, \hat{\Sigma})$ and take the $(1-\alpha)$ quantile of $\rho(\theta; \beta^*)$. This has no finite-sample validity, and in the boundary configuration it does certify falsely, at rate 0.157 at $\alpha = 0.5$ and 0.023 at $\alpha = 0.2$. Its

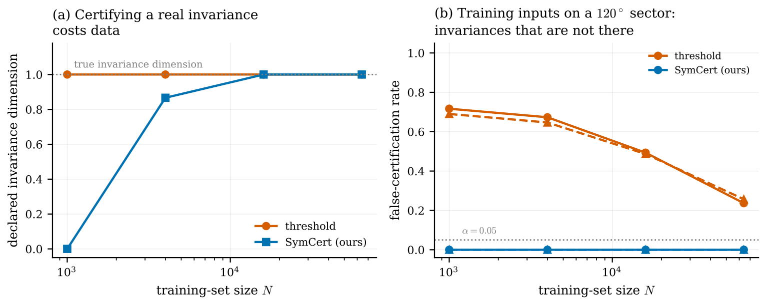

Fig. 7 Invariances of a learned model rather than of a dynamical system. **(a)** With training inputs covering the disc, both rules eventually recover the one-dimensional rotational invariance of the target; certification costs roughly sixteen times more data at this noise level. **(b)** With training inputs confined to a 120° sector, the fitted model is unconstrained elsewhere, and the threshold rule declares invariances that the target does not have. Solid lines are the rotation-invariant target, dashed the target with no invariance; the two are almost indistinguishable, which is the point, since the threshold rule fails the same way whether or not there is anything to find. SymCert declines in both cases.

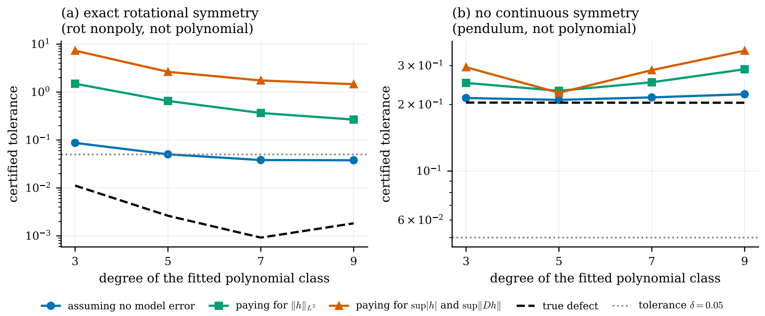

Fig. 8 Certifying a system that is not in any polynomial class. Ignoring the approximation error (blue) gives a bound that is not valid; paying for it in supremum norm (orange) is valid but vacuous; paying for it in L^2 against a richer polynomial class (green) is valid and several times tighter, and improves as the fitted class grows, though on the left-hand system it is still above the tolerance at degree 9, so the exact rotational symmetry that is really there cannot be certified honestly at this sample size. The dashed line is the true defect, obtained by Monte-Carlo integration of the analytic bracket; the dotted line is the tolerance $\delta = 0.05$.

intervals are 9% to 18% narrower than ours. That is the measurable price of exactness on this problem: between a tenth and a fifth of the width.

5.6 Departures from the assumptions

Table 3 reports coverage under errors that are Student-t_4, Laplace, or heteroskedastic, with the exact Gaussian ellipsoid and with a sandwich covariance (White, 1980), and under trajectory data where the rows are strongly dependent. Coverage never falls below nominal in any of the 48 settings, and no false certification occurs in any of them. The certificate is not fragile to the shape of the error distribution, because it depends on the errors only through a confidence ellipsoid for a least-squares estimator.

error distribution	design	worst coverage exact Gaussian	sandwich	worst false certification
Gaussian	i.i.d. design	1.000	1.000	0.000
Gaussian, heteroskedastic	i.i.d. design	1.000	1.000	0.000
Laplace	i.i.d. design	1.000	1.000	0.000
Student t_4	i.i.d. design	1.000	1.000	0.000
Gaussian	trajectories, exact states	—	1.000	0.000
Gaussian	trajectories, noisy states	—	1.000	0.000

Table 3 Coverage of the certificate under departures from the homoskedastic Gaussian model, at nominal 0.95, over 24 design settings and 24 trajectory settings. The certificate depends on the errors only through a confidence ellipsoid for a least-squares estimator, and is correspondingly insensitive to their shape and to dependence between rows. The last block is the case the guarantee does *not* cover: with noisy states and differenced derivatives the regressors carry error, and coverage held here only because the resulting bias happens to inflate the defect (Section 5.6).

The assumption that does bind is that the *states* are observed without error. We test it on trajectory data, twelve solution segments per replication so that the rows are strongly dependent as well, in two modes: exact states with noisy right-hand sides, and noisy states with derivatives taken by central differences. With exact states the certified dimension on the Hopf system is 1.00 at state-noise level 0.01 and 0.98 at 0.05. With noisy states it is 0.07 and 0.09: the errors-in-variables bias attenuates the fitted coefficients, which inflates the defect and destroys detection. Coverage held at 1.0 in all 24 trajectory settings, but that is a property of this particular attenuation acting in the conservative direction, not a guarantee. For that regime the honest options are the weak or integral formulation of model discovery (Messenger & Bortz, 2021), ensemble resampling of the fit (Fasel, Kutz, Brunton, & Brunton, 2022), or an explicit errors-in-variables treatment; we do not claim validity there.

Figure 8 completes the picture for genuinely non-polynomial systems. On a non-polynomial planar flow that has an exact rotational symmetry, a degree-5 fit gives a certified tolerance of 0.050 if model error is ignored, 0.65 if it is paid for in L^2 against a degree-9 class, and 2.65 if it is paid for in supremum norm; the last two are both vacuous. Raising the fitted degree to 9 shrinks the measured relative L^2 model error to 0.5% and the certified tolerance to 0.27, still too wide to be useful. The reason is the derivative amplification noted in Section 3.3: the bracket differentiates, so the constant converting a bound on $\|h\|$ into a bound on $\|[\xi, h]\|$ grows with the degree of the class h is allowed to occupy, and it grows faster than the approximation error falls.

On the damped pendulum, which has no continuous symmetry, the same accounting is cheap (true defect 0.204, certified bounds between 0.21 and 0.35 across all four degrees and both forms) because a degree-5 polynomial reproduces sin on the target domain to a relative L^2 error of 0.07%. Note also that the model error is not monotone in the degree: at degree 9 the fit is ill-conditioned enough that both η_0 and η_2 increase again. The practical rule of Section 3.3 therefore needs one qualification: enlarge the model class until a lack-of-fit test is satisfied, and no further.

n	P	Q	true dim.	certified tolerance (median)			CPU-s / cert.
				N=800	N=3200	N=12800	
2	4	6	2	0.0072	0.0036	0.0017	0.015
3	9	12	3	0.0110	0.0054	0.0027	0.017
4	16	20	4	0.0122	0.0069	0.0031	0.019
5	25	30	5	0.0194	0.0097	0.0048	0.024
6	36	42	6	0.0210	0.0108	0.0050	0.031

Table 4 Scaling in the state dimension, on linear systems whose symmetry algebra is the commutant of the coefficient matrix. The candidate search space $P = n^2$ grows quadratically with no cost to validity; what costs is the model class Q. False certification was zero in all 20 cells.

5.7 Scaling with the state dimension

The state dimension enters twice: through the number of model coefficients Q, which sets the radius of the confidence ellipsoid, and through the number of candidate generators P, which sets how many directions are searched. Only the first should matter, and Table 4 bears that out. On linear systems in $n = 2, \ldots, 6$ built from rotation blocks, whose symmetry algebra is the commutant of the coefficient matrix and has dimension n, false certification is zero in all 20 cells while the search space $P = n^2$ grows from 4 to 36. What the growth costs is width: at $N = 12800$ the certified tolerance rises from 0.0017 at $n = 2$ ($Q = 6$) to 0.0050 at $n = 6$ ($Q = 42$), a factor 2.9 against the $\sqrt{Q}$ prediction of 2.6. The whole n-dimensional algebra is recovered in every replication from $N = 800$ at $n = 4$ and from $N = 3200$ at $n = 6$. One certificate costs between 0.015 and 0.031 CPU-seconds across this range, dominated by the $O(Q^3)$ eigendecomposition inside the trust-region solves.

5.8 A case study on real measurements

The Hudson's Bay Company lynx and hare pelt records for 1900–1920 are the canonical real data for two-dimensional predator–prey model discovery: 21 annual counts, from which central differences give 19 usable state–derivative pairs. Fitting a quadratic vector field and running the pipeline gives the plug-in defect spectrum (0.18, 0.33, 0.68, 1.17, 1.31, 1.40), an extrapolation factor $\sqrt{\kappa\kappa'} = 58$, and a smallest certified tolerance of 2.3, larger than the largest value the defect can take. **With nineteen observations, no symmetry claim of any kind is supported.** The plug-in spectrum nonetheless *looks* informative, and at a tolerance of 0.3, which is not an unreasonable choice, a threshold rule would report a one-dimensional symmetry algebra.

We present this as a case study rather than as a validity claim: derivatives differenced from noisy annual counts put error in the regressors, which (Section 5.6) is the departure our guarantee does not cover.

5.9 Invariances of a learned model

The same machinery applies where F is a learned function and ξ generates an invariance: the Lie derivative becomes $\nabla f \cdot \xi$, and the defect becomes the root-mean-square cosine of the angle between the model's gradient and the generator, again a ratio of norms of linear images of the coefficients. We fit degree-6 polynomial regressions to noisy samples of three families of planar target: a rotation-invariant one, three that are invariant only up to a controlled amount, and one that is not. We then ask which invariances *of the target* may be declared. Figure 7 reports the result. With the training inputs covering the disc, both the threshold rule and SymCert recover the one-dimensional invariance once N is large enough, SymCert requiring roughly 16 times more data at noise level 0.05. With the training inputs covering a 120° sector, the threshold rule declares invariances that the target does not have, at rates up to 0.76, for both an invariant and a non-invariant target; SymCert never does.

This is a second and distinct coverage mechanism, and it completes the picture of Section 5.2. There the *integral* was extrapolated: the Gram matrix computed on the data stood in for the one on the target domain, and κ measured the gap. Here the integral is exact and the *model* is extrapolated: outside the sector the fit is unconstrained, and it drifts toward functions that happen to look invariant. No diagnostic on the states alone can see this. SymCert does, because the same lack of constraint that lets the fit drift also inflates $\hat{\Sigma}$ in exactly those directions, and the certificate is then too wide to clear any useful tolerance.

6 Related work

Linear-algebraic symmetry discovery.

The idea that continuous symmetries are a nullspace runs through work on point clouds (Cahill et al., 2023), conserved quantities (Kaiser et al., 2018), trained networks (Gruver et al., 2023; Moskalev et al., 2022) and dynamic mode decomposition (Baddoo, Herrmann, McKeon, Kutz, & Brunton, 2023); Otto et al. (2025) unify these, extend them to nonlinear group actions on manifolds and to sections of vector bundles, and add convex penalties that promote symmetry during training. Their conclusion names our problem: it will be important "to study the perturbative effects of noisy data in algorithms to discover and promote symmetry, with the goal of understanding the effects of problem dimension, noise level, and amount of data." We take up that question and answer it with confidence statements rather than with a perturbation analysis, because a perturbation bound alone does not tell a user what to declare. The Weyl-inequality baseline in our experiments is the perturbation answer, and it is valid but never certifies.

Optimisation-based discovery.

An alternative line searches for generators by nonlinear optimisation or adversarial training (Dehmamy, Walters, Liu, Wang, & Yu, 2021; Desai, Nachman, & Thaler, 2022; Forestano et al., 2023; Krippendorf & Syvaeri, 2020; Liu & Tegmark, 2022; Miao & Rao, 2007; Rao & Ruderman, 1999; Yang, Dehmamy, Walters, & Yu, 2024; Yang, Walters, Dehmamy, & Yu, 2023), and a related line learns invariances as hyperparameters through augmentation or marginal likelihood (Benton, Finzi, Izmailov, & Wilson, 2020; Immer, van der Ouderaa, Rätsch, Fortuin, & van der Wilk, 2022; Romero & Lohit, 2022; van der Wilk, Bauer, John, & Hensman, 2018). These methods return a point estimate of a generator with no error control; the question we ask, when may a discovered generator be declared and at what tolerance, applies to them too. Our machinery transfers whenever the Lie derivative is bilinear in the generator coordinates and the model coefficients, which covers any model that is linear in its parameters (basis-function regression, kernel and random-feature models, the last layer of a network). For a fully nonlinear model the ellipsoid must be replaced by a confidence region obtained some other way, and the trust-region step by an optimisation over it; the logic of Proposition 7 is unchanged.

Using a discovered symmetry.

Once a symmetry is in hand it is enforced, or promoted, or used to augment data: as hard linear constraints and convex penalties (Otto et al., 2025), as physics-informed losses (Akhound-Sadegh, Perreault-Levasseur, Brandstetter, Welling, & Ravanbakhsh, 2023; Raissi, Perdikaris, & Karniadakis, 2019), as data augmentation (Brandstetter, Welling, & Worrall, 2022), or inside equation discovery itself (Yang, Rao, Dehmamy, Walters, & Yu, 2024). Everything downstream inherits the reliability of the discovery step, which is the argument for making that step honest: a symmetry enforced on the strength of a small singular value is a modelling error that subsequent fitting cannot undo.

Approximate symmetry.

That real systems are only approximately symmetric is well recognised in modelling (Romero & Lohit, 2022; Wang, Walters, & Yu, 2022), and Otto et al. (2025) ask explicitly whether their framework can be used to "find elements that are nearly symmetric, and to bound the degree of asymmetry". Definition 1 is such a bound, and SymCert supplies its confidence statement.

Statistics.

The equivalence-testing formulation is standard in settings where the interesting hypothesis is the null (Berger & Hsu, 1996; Schuirmann, 1987; Wellek, 2010); our contribution is not the idea but the observation that symmetry discovery is one of those settings, together with a closed form for this particular functional. Rank tests (Cragg & Donald, 1997; Kleibergen & Paap, 2006; Robin & Smith, 2000) attack the same nullspace-dimension question and place symmetry in the null; we include one as a baseline and quantify the consequence. The selection issue is the one addressed by post-selection and split-sample inference (Andrews et al., 2024; Berk et al., 2013; Chernozhukov et al., 2018; J.D. Lee et al., 2016; Meinshausen et al., 2009); here simultaneity over the whole candidate space dissolves it more cheaply than splitting, which we verify empirically. Because a single event covers all generators at once, no multiplicity correction of the Benjamini and Hochberg (1995) kind is needed, and none would help, since the family being controlled is uncountable. Finally, the shape of the problem (many moment conditions, a low-dimensional parameter, and a test of whether they can all be satisfied) is the shape of a test of overidentifying restrictions (Hansen, 1982), with the difference that here the restriction of interest is the one we would like to accept rather than reject.

7 Discussion

What changes in practice.

Fix the region ν where the symmetry is supposed to hold, and a tolerance δ on the relative defect. Fit a model class large enough that a lack-of-fit test is satisfied. Compute $U(\theta)$; certify what falls below δ; report κ so the reader can see how much of the target domain the data actually visited. Report U itself rather than a binary verdict: it is the smallest tolerance the data support, and the honest analogue of the singular value that would otherwise be reported.

Limitations.

- *The model class is the binding assumption.* Validity requires the truth to lie in the fitted class, or an assumed bound on how far outside it lies. The certificate cannot rescue this. A fitted model can be more symmetric than the system it models, and no analysis of that model can detect the difference. Our remedies are a lack-of-fit gate, enlarging the class, and a sensitivity analysis in the size of the model error; none is a substitute for thought about the class.
- *Errors-in-variables.* When the states themselves are noisy and derivatives are differenced, the regressors are noisy, the estimator is biased, and the confidence ellipsoid is not valid. In our runs the bias acted conservatively, but we do not claim validity there; weak-form model discovery (Messenger & Bortz, 2021) is the natural route.
- *Conservatism.* The certificate over-covers, its interval containing the truth in every replication even at nominal 0.5, at a measured cost of 9% to 18% in width relative to a calibration-targeting bootstrap (Section 5.5) and a factor 8.1–8.8 relative to the Le Cam lower bound (Section 5.3), part of which is the looseness of that bound rather than of the certificate. Closing the measurable part of the gap, by exact quantiles for the ratio functional in place of separate bounds on numerator and denominator, is the most concrete direction for improvement.
- *Scope.* We treat connected symmetry groups whose generators lie in a prescribed finite-dimensional class, and moderate problem sizes ($n \leq 6$, $Q \leq 105$, $P \leq 36$). Discrete symmetries, and symmetries visible only after a nonlinear change of coordinates (Liu & Tegmark, 2022; Yang, Dehmamy, et al., 2024), are outside the framework as they are outside the Lie-algebraic setting generally.
- *Mostly simulated systems.* The benchmarks are simulated deliberately: the claims are about error rates, and measuring an error rate requires knowing the truth exactly, which a real system does not provide. We include one real data set

(Section 5.8), where the honest answer turns out to be that nothing can be concluded. That is informative about the method, but not a substitute for a setting in which symmetries are both real and known.

The wider point.

Symmetry discovery is an instance of a pattern that recurs whenever a scientific claim takes the form "this quantity is zero": a rank, a conservation law, a conditional independence, a null effect. In each, an estimate is compared to a tolerance, and the tolerance is asked to carry statistical weight it cannot bear. The remedy is the same each time: state the tolerance in interpretable units, and require the data to *exclude* the alternative instead of merely failing to detect it. Whether that remedy is cheap depends on the functional; for the Lie derivative it turns out to cost two trust-region subproblems.

Supplementary information. No supplementary files accompany this article. The software, the raw result files and the scripts that regenerate every figure, table and quoted number are in the repository named under Data availability below.

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Declarations

- **Funding.** No funds, grants or other support were received for conducting this study.
- **Competing interests.** The author has no competing interests to declare that are relevant to the content of this article.
- **Ethics approval and consent to participate.** Not applicable. This work involves no human participants, no animals and no personal data.
- **Consent for publication.** Not applicable.
- **Data availability.** All data analysed in this study are either generated by the released simulation code or, in the single case of real measurements, taken from a published table. The Hudson's Bay Company lynx and hare pelt counts for 1900–1920 used in Section 5.8 are the values distributed with the Stan example models (`knitr/lotka-volterra/hudson-bay-lynx-hare.csv`, https://github.com/stan-dev/example-models) and are reproduced verbatim in `experiments/exp12_realdata.py`. All raw result files are in the `results/` directory of https://github.com/Kendiukhov/certified-symmetry-discovery.
- **Materials availability.** Not applicable.
- **Code availability.** The complete implementation, the experiment scripts, the figure and table generators and the test suite are released under the MIT licence at https://github.com/Kendiukhov/certified-symmetry-discovery.
- **Author contribution.** I.K. is the sole author and is responsible for the conception, the theory, the implementation, the experiments and the manuscript.

Appendix A Proofs

Lemma 2

Write $u = DF\,\xi$ and $v = D\xi\,F$ as elements of $L^2(\nu)$. Then $\rho^2 = \|u-v\|^2/(\|u\|^2+\|v\|^2)$. The parallelogram-type inequality $\|u-v\|^2 \le 2(\|u\|^2+\|v\|^2)$ gives $\rho \le \sqrt{2}$, with equality iff $u = -v$; $\rho = 0$ iff $u = v$ in $L^2(\nu)$, i.e. $[\xi, F] = 0$ ν-almost everywhere, and since both are polynomials this holds on the support of ν; that this is equivalent to the flow of ξ mapping solutions to solutions is standard (J.M. Lee, 2013, Ch. 9). Under $\xi \mapsto c\xi$ both u and v scale by c; under $F \mapsto c'F$ both scale by c'; the ratio is invariant in each case, hence also under any invertible linear reparameterisation of the candidate class, because ρ depends on θ only through the field ξ_θ. For $x = cQz$ with Q orthogonal, the pullbacks are $\tilde{F}(z) = c^{-1}Q^\top F(cQz)$ and likewise for ξ, so $\tilde{u} = c^{-1}Q^\top u \circ (cQ)$ and similarly for v; the factor $c^{-1}Q^\top$ is a scalar multiple of an isometry and cancels in the ratio once ν is transported. $\square$

Lemma 6

DF_β and F_β are linear in β, and ξ_θ and $D\xi_\theta$ are linear in θ, so each of $DF_\beta\,\xi_\theta$ and $D\xi_\theta F_\beta$ is a polynomial vector field whose coefficient vector is bilinear in (θ, β). Collecting coefficients over the monomial basis of the residual space and multiplying by the symmetric square root of the monomial Gram matrix of ν (which exists because the Gram matrix is positive semi-definite) turns the $L^2(\nu)$ norms into Euclidean norms, giving $M_\theta^{(1)}$ and $M_\theta^{(2)}$; L_θ is their difference by (1). Assembling $C(\beta) = A(\beta)^\top A(\beta)$ with $A(\beta)_{\cdot p} = L_{e_p}\beta$ and $D(\beta)$ analogously, $\rho^2 = \theta^\top C\theta/\theta^\top D\theta$, whose minimum over $\theta \ne 0$ is the smallest generalised eigenvalue of (C, D) whenever $D \succ 0$. $\square$

Proposition 7

On the event $E = \{\beta^\star \in \mathcal{B}\}$ we have, for every θ, $\|L_\theta\beta^\star\| \le \sup_{\beta\in\mathcal{B}} \|L_\theta\beta\|$ and $\|M_\theta^{(j)}\beta^\star\| \ge \inf_{\beta\in\mathcal{B}} \|M_\theta^{(j)}\beta\|$ for $j = 1, 2$, so $\rho_\nu(\xi_\theta; F_{\beta^\star}) \le U(\theta)$; the lower bound is symmetric. Since $\mathbb{P}(E) \ge 1-\alpha$ and E does not depend on θ, the statement holds simultaneously. For (6): on E, any ξ with $U(\xi) \le \delta$ has $\rho_\nu(\xi; F_{\beta^\star}) \le \delta$, whatever rule produced it; so the bad event is contained in E^c. $\square$

Proposition 8

Writing $\mathcal{B} = \{\hat\beta + \hat\Sigma^{1/2}u : \|u\| \le R\}$, each extremum is max or min of $\|a + Bu\|$ over $\|u\| \le R$ with $a = L_\theta\hat\beta$ and $B = L_\theta\hat\Sigma^{1/2}$. Squaring, this is the optimisation of $u^\top Qu + 2c^\top u$ over a ball with $Q = B^\top B \succeq 0$ and $c = B^\top a$: the trust-region subproblem, whose solutions satisfy $(Q \mp \mu I)u = \mp c$ with the multiplier fixed by the secular equation $\sum_i \tilde{c}_i^2/(\mu \mp \lambda_i)^2 = R^2$ in the eigenbasis of Q, monotone in μ and solvable by bisection; the hard case (c orthogonal to the extreme eigenspace) is handled by adding a component in that eigenspace (Conn, Gould, & Toint, 2000; Gander et al., 1989; Moré & Sorensen, 1983). For the exactness of the region: in model (3), $\hat\beta - \beta^\star \sim N(0, \sigma^2\Sigma_0)$ with $\Sigma_0 = I_n \otimes (\Phi^\top\Phi)^{-1}$ and $\mathrm{RSS}/\sigma^2 \sim \chi^2_{\mathrm{dof}}$ independent of $\hat\beta$, so $\|\hat\Sigma^{-1/2}(\hat\beta - \beta^\star)\|^2/Q$, with $\hat\Sigma = \hat\sigma^2\Sigma_0$, is exactly $F_{Q,\mathrm{dof}}$ distributed. $\square$

Proposition 9

Fix ξ, and let M_ξ stack $M_\xi^{(1)}$ and $M_\xi^{(2)}$. Choose Δ attaining $s_{\max}(\xi)$, normalised so $\|F_\Delta\|_{L^2(\mu_N)} = 1$ where μ_N is the empirical design measure, and set $\beta^t = \beta^0 + t\Delta$. Since $L_\xi\beta^0 = 0$, $\|L_\xi\beta^t\| = t\, s_{\max}$ and $\|M_\xi\beta^t\| \le \|M_\xi\beta^0\| + t\|M_\xi\Delta\|$, so $\rho_\nu(\xi; F_{\beta^t}) \ge \delta$ for $t \ge t_\delta = \delta\|M_\xi\beta^0\|/(s_{\max} - \delta\|M_\xi\Delta\|)$. The Kullback–Leibler divergence between the two data distributions is $\mathrm{KL} = \frac{Nt^2}{2\sigma^2}\|F_\Delta\|^2_{L^2(\mu_N)} = Nt^2/(2\sigma^2)$, so by Pinsker's inequality the total variation distance is at most $t\sqrt{N}/(2\sigma)$. For any test φ, $\mathbb{P}_{\beta^0}(\varphi = 1) \le \mathbb{P}_{\beta^t}(\varphi = 1) + \mathrm{TV} \le \alpha + t\sqrt{N}/(2\sigma)$ for any t slightly above t_δ; taking $t \downarrow t_\delta$ and expanding $t_\delta = \delta\|M_\xi\beta^0\|/s_{\max} + O(\delta^2)$ gives the claim. The argument is the standard two-point (Le Cam) method (Tsybakov, 2009, Ch. 2). □

Proposition 10

Both $\|r\|^2_{L^2(\nu)}$ and $\|r\|^2_{L^2(\mu)}$ are quadratic forms in the coefficient vector c of $r \in \mathcal{R}$, equal to $c^\top G_\nu c$ and $c^\top G_\mu c$; their ratio is a generalised Rayleigh quotient whose maximum is $\lambda_{\max}(G_\mu^{-1/2} G_\nu G_\mu^{-1/2})$, attained at the corresponding eigenvector, which is a legitimate element of $\mathcal{R}$. The defect statement follows by applying the bound to the numerator and its reverse to each denominator block. If G_μ is singular with null vector $c \ne 0$ then $\|r_c\|_{L^2(\mu)} = 0$ while $\|r_c\|_{L^2(\nu)} > 0$ (as $G_\nu \succ 0$ for a measure with a density on an open set), so $\kappa = \infty$; conversely $\kappa = \infty$ forces G_μ to be singular. □

Appendix B Construction of the tensors

Represent a polynomial vector field on $\mathbb{R}^n$ by an $n \times m$ array of coefficients over the monomials of total degree at most D. Multiplication and differentiation of monomials are index tables computed once. The bracket (1) of two such fields is then exact integer/rational linear algebra. Fixing the candidate basis $\{\zeta_p\}$ and the model basis $\{x^a e_i\}$ and computing $[\zeta_p, x^a e_i]$ for every pair gives a three-index tensor $T \in \mathbb{R}^{K\times P\times Q}$ with $\mathrm{coef}([\xi_\theta, F_\beta]) = T\cdot\theta\cdot\beta$, and likewise $T^{(1)}, T^{(2)}$ for the two individual terms, with $T = T^{(1)} - T^{(2)}$. Multiplying the first index by $I_n \otimes g_\nu^{1/2}$, where g_ν is the monomial Gram matrix of ν, converts Euclidean norms of contracted vectors into $L^2(\nu)$ norms of the corresponding fields. Then $L_\theta = T_w \cdot \theta$ and $A(\beta) = T_w \cdot \beta$.

g_ν is available in closed form for the uniform measure on a box, the Gaussian, and the uniform measure on a ball or annulus (the last two by separating radial and angular moments), so no quadrature error enters. For the *data* measure we substitute the empirical Gram matrix of the same monomial basis; the difference between the two is exactly what Proposition 10 measures.

For the refutation bound we need $\min_{\theta\ne 0}\rho_\nu(\xi_\theta; F_{\beta^\star})$ from below. On the event $\{\beta^\star \in \mathcal{B}\}$, $\sigma_{\min}(A(\beta^\star)) \ge \sigma_{\min}(A(\hat\beta)) - \sup_{\beta\in\mathcal{B}}\|A(\beta - \hat\beta)\|_F$ and $\sigma_{\max}(M(\beta^\star)) \le \sigma_{\max}(M(\hat\beta)) + \sup_{\beta\in\mathcal{B}}\|M(\beta - \hat\beta)\|_F$, and each supremum is again a norm of a linear image of β, hence another trust-region maximum. The ratio of the two bounds is the reported lower bound. The subspace certificate is the same construction with A and M restricted to the subspace.

Appendix C Baselines

Fixed threshold.

Declare the span of all generalised eigenvectors of $(C(\hat{\beta}), D(\hat{\beta}))$ whose plug-in defect is at most τ. Sweeping τ traces the whole family of threshold rules, including the choice $\tau = \delta$ that matches our tolerance.

Eigengap.

Declare d maximising the relative gap $\rho_{(d+1)}/\rho_{(d)}$ over $d \geq 1$. This rule cannot express "no symmetry", which is one reason its error rate is high; we report it because it is widely used, and we also report the threshold family, which can.

Weyl perturbation certificate.

Bound $\lambda_k(C(\beta^\star)) \leq \lambda_k(C(\hat{\beta})) + \|C(\beta^\star) - C(\hat{\beta})\|$ with the perturbation bounded by $2\|A(\hat{\beta})\|\,\|\Delta A\|_F + \|\Delta A\|_F^2$ and $\|\Delta A\|_F$ maximised over the same confidence ellipsoid, then divide by a lower bound on $\lambda_{\min}(D)$. This is the natural singular-value perturbation answer (Bhatia, 1997; Davis & Kahan, 1970; Yu, Wang, & Samworth, 2015), the same route one would take with matrix concentration (Tropp, 2012) if the Gram matrix were estimated by averaging over samples rather than integrated exactly; it is valid, and it is loose because it spends one global perturbation budget on every direction at once instead of bounding the ratio direction by direction.

Rank test.

For each d, test H_0: the d smallest plug-in directions span an exact nullspace, using the squared $L^2(\nu)$ norm of the restricted bracket referred to a Welch–Satterthwaite approximation of its null distribution (a weighted sum of χ_1^2 variables), and report the largest d not rejected. This is a Wald-type rank test in the spirit of Robin and Smith (2000) and Kleibergen and Paap (2006).

Split significance test.

Select the smallest-defect direction on one half of the data; on the other half test H_0: that direction is an exact symmetry, by the same Wald statistic; declare a symmetry when the test fails to reject. The selection is honest and the test is valid; the declaration is not.

Appendix D Reproducibility

All experiments are deterministic given the seeds recorded in the code. The package, the experiment scripts, the figure scripts and the raw result files are released together at https://github.com/Kendiukhov/certified-symmetry-discovery. Every table body in the manuscript is written by `paper/make_tables.py` directly from those files rather than transcribed, and every number that appears in the running text is asserted against them by `paper/check_numbers.py`, which verifies 31 claims and fails if any has drifted.

`results/compute_log.json` records the CPU and wall time of the most recent run of each experiment; the recorded total for one complete pass of the suite is given in the introduction. Wall-clock times in that file are not meaningful, because the machine was shared with unrelated jobs throughout, and the CPU times should be read as the cost of one pass, not as the total cost of the work, since several experiments were re-run during development.

The correctness of the mathematical core is checked by a test suite that verifies: the bracket against finite differences; the closed-form monomial moments of all four target measures against Monte-Carlo integration; the defect against Monte-Carlo evaluation of its defining integrals; the exact symmetry algebra of linear systems against the dimension of the commutant computed by an independent route; the trust-region extrema against brute-force search over the ball; the Welch–Satterthwaite approximation against simulation; and the empirical coverage of the certificate against its nominal level.

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